

◆◆ Topic-Wise Question List

1. Gauss's Law & Electrostatics

1. State and explain **Gauss's Law**.
2. What is meant by a **Gaussian surface**? Give examples.
3. Deduce **Coulomb's Law** from **Gauss's Law**.
4. Derive the expression for **electric potential at a distance r from an isolated point charge q** .
5. An **infinite-length rod** has a linear charge density λ . Calculate the **electric field intensity** at a distance from the rod.
6. Write the **expression of electric field intensity** due to a **ring of charge q and of radius a** .
7. Deduce an **expression for the electric field intensity** due to a **long straight, uniformly charged wire**.

2. Coulomb's Law & Electric Force

8. **Define Coulomb's Law**. Derive it from **Gauss's Law**.
9. Two charges, $q_1 = +5\mu\text{C}$ and $q_2 = -3\mu\text{C}$, are placed **0.4 m apart** in a vacuum.
 - (a) Find the **electrostatic force** between them.
 - (b) Find the **electric field intensity** at a point **0.2 m from q_1** along the line joining the two charges.
 - (c) Find the **electric potential** at the same point due to both charges.
10. Two point charges, $q_1 = +3\mu\text{C}$ and $q_2 = -2\mu\text{C}$, are placed **0.5 m apart in vacuum**. Find:
 - (a) The **magnitude of the electrostatic force** between them.
 - (b) The **electric field intensity** at the midpoint.
 - (c) The **direction of the electrostatic force** on each charge.

3. Electric Dipole & Potential

11. **Define electric dipole, dipole moment, and quantization of charge**.
12. Derive the **expression for electric field intensity at a point on the perpendicular bisector of a dipole**.
13. Calculate the **electric potential for an electric dipole**.
14. Derive an **expression for electric potential due to a charged disk**.
15. Write the **expression for torque and potential energy of an electric dipole** in a uniform electric field.

4. Capacitance & Energy Storage

16. What is capacitance? Derive the expression for energy stored in a charged capacitor. 17. What is a parallel plate capacitor? Derive an expression for its capacitance. 18. Calculate the capacitance of a parallel plate capacitor with plate area A and separation d . 19. Define dielectric constant and explain its significance. 20. What are the three electric vectors in a dielectric medium? 21. Calculate the equivalent capacitance for series and parallel capacitors, where:

- $C_1=5\mu\text{F}$, $C_2=8\mu\text{F}$, $C_3=10\mu\text{F}$
- Capacitor series parallel math problem list.

5. Kirchhoff's Laws & Electric Circuits

22. State and explain Kirchhoff's Voltage Law (KVL). 23. State and explain Kirchhoff's Current Law (KCL). 24. Define the terms:

- Current
- Current density
- Resistance & Resistivity
- Electric potential

6. Resistance & Conductivity

25. A rectangular carbon block has dimensions of $1.0\text{ cm} \times 1.0\text{ cm} \times 50\text{ cm}$.

- I. What is the resistance measured between the two square ends?
II. What is the resistance measured between two opposing rectangular faces? The resistivity of carbon at 20°C is $3.5 \times 10^{-5}\text{ ohm-meter}$.

$$R = \rho \frac{L}{A}$$

Ohms law Math

Formula math